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Inventor(s)	Olson; John et al.

Wall mounted HVAC system

Abstract

A wall mounted HVAC system includes a housing having an internal cosmetic panel, a holder supporting an air filter, and at least one external vent, the wall mounted HVAC system, further having a 48 Volt DC electric compressor, an evaporator coil, a condenser coil, a 48 Volt DC dual blower system, an over-temperature protection system, and a thermostatic expansion valve. The wall mounted HVAC system is configured to attach to a building structure and may be used when the building structure does not have readily available power from a generator or electrical grid.

Inventors:	Olson; John (Fort Worth, TX), Western; George (Jacksonville, TX)
Applicant:	Victory Climate Systems, LLC (Fort Worth, TX)
Family ID:	1000007385691
Assignee:	Victory Climate Systems, LLC (Fort Worth, TX)
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Primary Examiner: Trpisovsky; Joseph F

Background/Summary

(1) This application claims the benefit of U.S. Provisional Patent Application No. 63/391,910, filed 25 Jul. 2022, titled “Wall Mounted Climate Control System,” and U.S. Provisional Patent Application No. 63/436,355, filed 30 Dec. 2022, titled “Wall Mounted HVAC System,” both of which are incorporated herein for all purposes.

BACKGROUND

1. Field of the Invention
- (1) The present application relates generally to wall mounted climate control systems.
2. Description of Related Art
- (2) There are many different types of air conditioning systems in the market today. Some of these systems are designed to be mounted on the wall of a structure to provide air conditioning to the interior of the structure. However, relatively few of these systems are designed to run on DC power, and most are of those are underpowered and inefficient.
- (3) Although great strides have been made in the area of wall mounted climate control systems, considerable shortcomings remain.

Description

DESCRIPTION OF THE DRAWINGS

- (1) The novel features believed characteristic of the application are set forth herein. However, the application itself, as well as a preferred mode of use, and further objectives and advantages thereof, will best be understood by reference to the following detailed description when read in conjunction with the accompanying drawings, wherein:
- (2) FIG. 1 is a rear perspective view of a wall mounted climate control system according to one embodiment of the present application;
- (3) FIG. 2 is a detailed cut-away front perspective view of the wall mounted climate control system of FIG. 1;
- (4) FIG. 3 is a front view of the wall mounted climate control system of FIG. 1;
- (5) FIG. 4 is a right side view of the wall mounted climate control system of FIG. 1;
- (6) FIG. 5 is a left side perspective view of the wall mounted climate control system of FIG. 1;

- (7) FIG. 6 is a front perspective view of a wall mounted HVAC system according to a preferred embodiment of the present application;
- (8) FIG. 7 is a rear perspective view of the wall mounted HVAC system of FIG. 6;
- (9) FIG. 8 is a front view of the HVAC system of FIG. 6;
- (10) FIG. 9 is a left side view of the HVAC system of FIG. 6;
- (11) FIG. 10 is a right side view of the HVAC system of FIG. 6;
- (12) FIG. 11 is a front perspective, partially assembled, view of the HVAC system of FIG. 6, primarily showing the HVAC systems;
- (13) FIG. 12 is a rear perspective, partially assembled, view of the HVAC system of FIG. 6, primarily showing the HVAC systems;
- (14) FIG. 13 is a rear, partially assembled, view of the HVAC system of FIG. 6, primarily showing the HVAC systems;
- (15) FIG. 14 is a left side, partially assembled, view of the HVAC system of FIG. 6, primarily showing the HVAC systems; and
- (16) FIG. 15 is a right side, partially assembled, view of the HVAC system of FIG. 6, primarily showing the HVAC systems;
- (17) While the system and method of the present application is susceptible to various modifications and alternative forms, specific embodiments thereof have been shown by way of example in the drawings and are herein described in detail. It should be understood, however, that the description herein of specific embodiments is not intended to limit the application to the particular embodiment disclosed, but on the contrary, the intention is to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the process of the present application as defined herein and by any appended claims.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

- (18) Illustrative embodiments of the preferred embodiment are described below. In the interest of clarity, not all features of an actual implementation are described in this specification. It will of course be appreciated that in the development of any such actual embodiment, numerous implementation-specific decisions must be made to achieve the developer's specific goals, such as compliance with system-related and business-related constraints, which will vary from one implementation to another. Moreover, it will be appreciated that such a development effort might be complex and time-consuming but would nevertheless be a routine undertaking for those of ordinary skill in the art having the benefit of this disclosure.
- (19) In the specification, reference may be made to the spatial relationships between various components and to the spatial orientation of various aspects of components as the devices are depicted in the attached drawings. However, as will be recognized by those skilled in the art after a complete reading of the present application, the devices, members, apparatuses, etc. described herein may be positioned in any desired orientation. Thus, the use of terms to describe a spatial relationship between various components or to describe the spatial orientation of aspects of such components should be understood to describe a relative relationship between the components or a spatial orientation of aspects of such components, respectively, as the device described herein may be oriented in any desired direction.
- (20) Referring now to FIG. 1 in the drawings, a wall mounted climate control system **100** is illustrated. In this embodiment, system **100** is an electrical system and is configured to run on 48 Volt DC battery power. System **100** is particularly well suited to control the climate and air temperature inside small structures used to house telephony equipment. It will be appreciated that system **100** may be used with a wide variety of structures.
- (21) System **100** is configured to be a standalone air conditioning system for use on structures without power available from generator or normal electrical grids. For example, system **100** might be installed on a telephone data center building on a remote mountain top in Colorado, in which the building has its own solar charging and battery storage. In the preferred embodiment, power from

48 Volt batteries, a solar system, and/or other power sources, exist in the structure.

(22) System **100** includes a housing **101** and is configured to mount on the exterior of an existing structure using metal studs **102**. Housing **101** is held in place with an internal cosmetic panel **104** that incorporates a removable and replaceable air filter (not shown). The air filter is held in place by a holder **106**. Housing **101** includes one or more external vents with grills **105**, each grill **105** having an insect screen.

(23) System **100** further includes a 48 Volt DC electric compressor **110**, an aluminum plate fin evaporator coil **112**, a large face area tube fin condenser coil **114**, and a 48 Volt DC dual blower system **108**. Compressor **110** preferably includes a soft-start operation subsystem, which reduces inrush current when starting up. Evaporator coil **112** provides for maximum heat transfer of refrigerant to air. Condenser coil **114** reduces overall operating pressures. Large condenser air inlet grills **105** with insect screens prevent pest and nuisance power interruption. Dual blower system **108** provides sufficient airflow to maintain interior temperatures when called upon for operation. System **100** also includes two 48 Volt DC fans **116a** and **116b** that operate in parallel to provide redundant air flow for operation of condenser coil **114**.

(24) System **100** further includes an over-temperature protection system **118** and a thermostatic expansion valve **120**. System **100** includes an alarm system, which produces an output signal in the event of operation error. The alarm system and other control system features may interface with customer standard service call procedures. System may be controlled manually at a control panel connected to system **100**, or may be controlled remotely by a variety of wired and/or wireless devices and networks.

(25) System **100** may include a heating subsystem to provide heat to the enclosure.

(26) System **100** is an improvement over existing products of similar size and capacity in at least the following categories: greater air flow for faster temperature pull-down; larger evaporator coil with higher capacity; and greater condensing capacity (dual condenser fans instead of one).

(27) Referring now to FIGS. **6-15** in the drawings, a wall mounted heating, ventilation, and air conditioning (HVAC) system **1000** according to the preferred embodiment of the present application is illustrated. System **1000** includes separate heating, air conditioning, and ventilations subsystems. System **1000** is preferably a 48 Volt DC system. System **1000** is housed by and carried by a frame support system **1001** and includes a rear panel **1003** that is disposed within the structure being heated and cooled by system **1000**. Rear panel **1003** includes multiple mounting apertures **1005**, a supply air gasket **1007**, a return air gasket **1009**, and one or more apertures **1011** for power and control signal entry.

(28) System **1000** includes an evaporator **1002**, a condenser **1004**, and a dual scroll blower **1006**, and a specialized, unique control system **1008**. System **1000** also includes a power and control signal termination box **1010**, a thermal expansion valve **1012**, a thermal expansion valve refrigerant line **1014**, a suction refrigerant line **1016**, a drain pan **1018**, an evaporator freeze thermostat **1020**, one or more drain tubes **1022**, and a discharge refrigerant line **1026**. A return air plenum **1028** is coupled to dual scroll blower **1006** and opens out of rear panel **1003**.

(29) In addition, system **1000** includes a positive temperature coefficient (PTC) heating element **1024**. PTC heating element is preferably a 48 Volt DC 700 Watt heating element. It will be appreciated that other sizes of DC heating elements, and other types of DC heating elements may be used. System **1000** also includes dual condenser fans **1030**, a refrigerant drier/filter **1032**, a binary pressure switch **1033** (high and low pressure), and a 48 Volt DC electric refrigerant compressor **1034**.

(30) System **1000** includes the following features: 36~60 VDC (nominal 48 VDC) operation. Aluminum automotive plate-fin evaporator. 2.5 ton rated automotive thermal expansion valve with radial O-ring interfaces. Automotive style reduced barrier crimped refrigerant hoses. Inline axial filter/drier with sight-glass. 48 Volt DC nominal 27 cc automotive style electric scroll compressor. Oversized copper tube and aluminum bluefin condenser. Automotive binary pressure switch (hi and

low pressure detection) installed on liquid line for detection of system refrigeration leaks or overpressure. Dual automotive 48 VDC condenser fans, derated to reduce noise of operation. Dual-scroll 48 VDC evaporator blower. 48 VDC 700 W positive thermal coefficient (P.T.C.) heater. Automotive style fixed temperature thermostat for preventing evaporator freezing. Temperature sensor for detecting loss of blower during PTC heater operation. System control board for HVAC compressor, blower, fans, and heater operation in response to customer control signal input. Also provides alarm feedback. Ambient temperature detection for mitigating nuisance alarms due to binary pressure switch operation at low temperature, in which a nuisance alarm is any alarm that is triggered unintentionally or mistakenly. Alarm system monitors condition of fuses and alerts if a fuse is failed. Condensate drain tube(s) contain partial plug to allow operation, but prevent insect intrusion. Case framework and building attachment mounts target operation in up to Zone 4 (hurricane and earthquake). Customer 48 Volt DC hookup via application specific busbar design. System **10000** is particularly useful in the following applications: Cell or radio tower “radio shacks.” Small off grid/remote shelters. Backyard “play houses” or sheds. Generic 48 Volt DC 65 Amp+ applications. Windmill generator control cabinets/sheds. Low vibration mobile 48 Volt DC. High vibration (with additional support) mobile 48 Volt DC. For example, military, e.g. MRAP, etc. (31) The particular embodiments disclosed above are illustrative only, as the application may be modified and practiced in different but equivalent manners apparent to those skilled in the art having the benefit of the teachings herein. It is therefore evident that the particular embodiments disclosed above may be altered or modified, and all such variations are considered within the scope and spirit of the application. Accordingly, the protection sought herein is as set forth in the description. It is apparent that an application with significant advantages has been described and illustrated. Although the present application is shown in a limited number of forms, it is not limited to just these forms, but is amenable to various changes and modifications without departing from the spirit thereof.

Claims

1. A wall mounted HVAC system, comprising: a housing configured for attachment to a building structure using at least one metal stud, the housing comprising: an internal cosmetic panel; a holder configured to support a replaceable air filter; and at least one external vent; a 48 Volt DC electric compressor having a soft-start operation subsystem reducing rush current during start up; an evaporator coil for maximizing heat transfer of refrigerant to air; a condenser coil; and a 48 Volt DC dual blower system configured to provide sufficient airflow for maintaining interior temperatures; an over-temperature protection system; and a thermostatic expansion valve; wherein the wall mounted HVAC system is configured to be a standalone air conditioning system on structures without readily available power from a generator or electrical grid.
